Warmup/Attendance: [Week 12: Friday 11/13/2026]

Full Name: SID:

Work with the people around you to try to solve these problems, which we'll discuss together once lecture begins. These are not graded for correctness, but you will turn this sheet in for attendance at the end of class.

1. What are the best/worst values for Confusion matrix, MCR, Precision, Recall?

Imagine your sample has $n = 180$, with 70 positive observations, and 110 negative ones.

2. Estimated probabilities

```
m1 <- glm(sex ~ body_mass_g, data = penguins, family = "binomial")
```

```
##  (Intercept)  body_mass_g
## -5.162541644  0.001239819
```

Write down the mathematical expression for the estimated probability that a penguin that weighs 4000 g is a *female*. Then write down the code to extract the same number.

3. What is the misclassification rate, Precision, and Recall of this model?

```
## # A tibble: 4 × 3
## # Groups:   sex [2]
##   sex    sex_hat     n
##   <fct>  <chr>   <int>
## 1 female female    109
## 2 female male       56
## 3 male   female     74
## 4 male   male       94
```